Scalability & Future Plan

Date	6 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – AI-Based Crop Recommendation System
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the **OptiCrop – AI-Based Crop Recommendation System** can be enhanced to support more users, improve prediction performance, strengthen application reliability, and introduce advanced agricultural features in future versions. It also presents the team's roadmap for expanding the application's capabilities beyond the current implementation.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	No user authentication system	Anyone can access the application without login.	High
2	No database integration to store prediction history.	Previous crop recommendations cannot be saved or retrieved.	Medium
3	Recommendations are based only on soil and environmental parameters	The system does not provide fertilizer suggestions, weather forecasting, or disease prediction.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports limited concurrent users	Deploy the application on a scalable cloud platform with load balancing and auto-scaling.
2	Data Storage	No database integration	Integrate MySQL or PostgreSQL to store user accounts and crop recommendation history.
3	Performance	Single Flask application	Optimize model loading, caching, and backend processing for faster crop recommendations.
4	Security	Provides crop recommendation only.	Integrate Weather API, fertilizer recommendation, crop disease detection, and yield prediction modules.

Future Roadmap:

Phase	**Planned Feature / Enhancement**	**Target Timeline**	**Expected Impact**
Phase 2	Add user authentication and account management.	2–3 Months	Secure access and personalized user experience.
Phase 3	Store crop recommendation history using a database.	3–4 Months	Enable users to view and manage previous recommendations
Phase 4	Integrate weather forecasting, fertilizer recommendation, crop disease detection, and crop yield prediction.	5–6 Months	Improve recommendation accuracy and provide a complete intelligent farming assistance platform.